



SOLAR SAGE AI

-“We care for your panels, so your future shines brighter”.

Problem



Efficiency Loss: Dust and dirt can reduce solar panel output by **20-40%** in dusty/arid regions



Manual cleaning wastes water and labor **1GW** solar farm house **12.5M**+liters/month



Frequent manual maintenance leads to **panel micro-cracks**

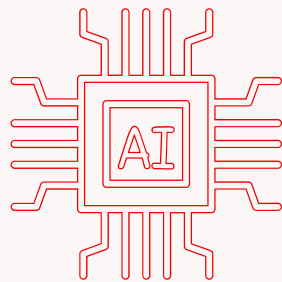


Blind Cleaning- Most solar plants still follow fixed, schedule-based cleaning, not condition-based cleaning

Solution



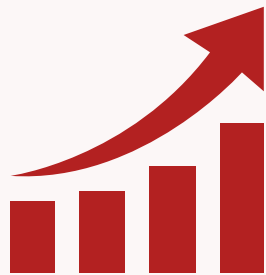
End-to-end perception-to-decision latency of <1 second, enabled by 100% on-edge processing with zero cloud dependency.



Uses Computer Vision & AI agents directly on the edge (no cloud lag).

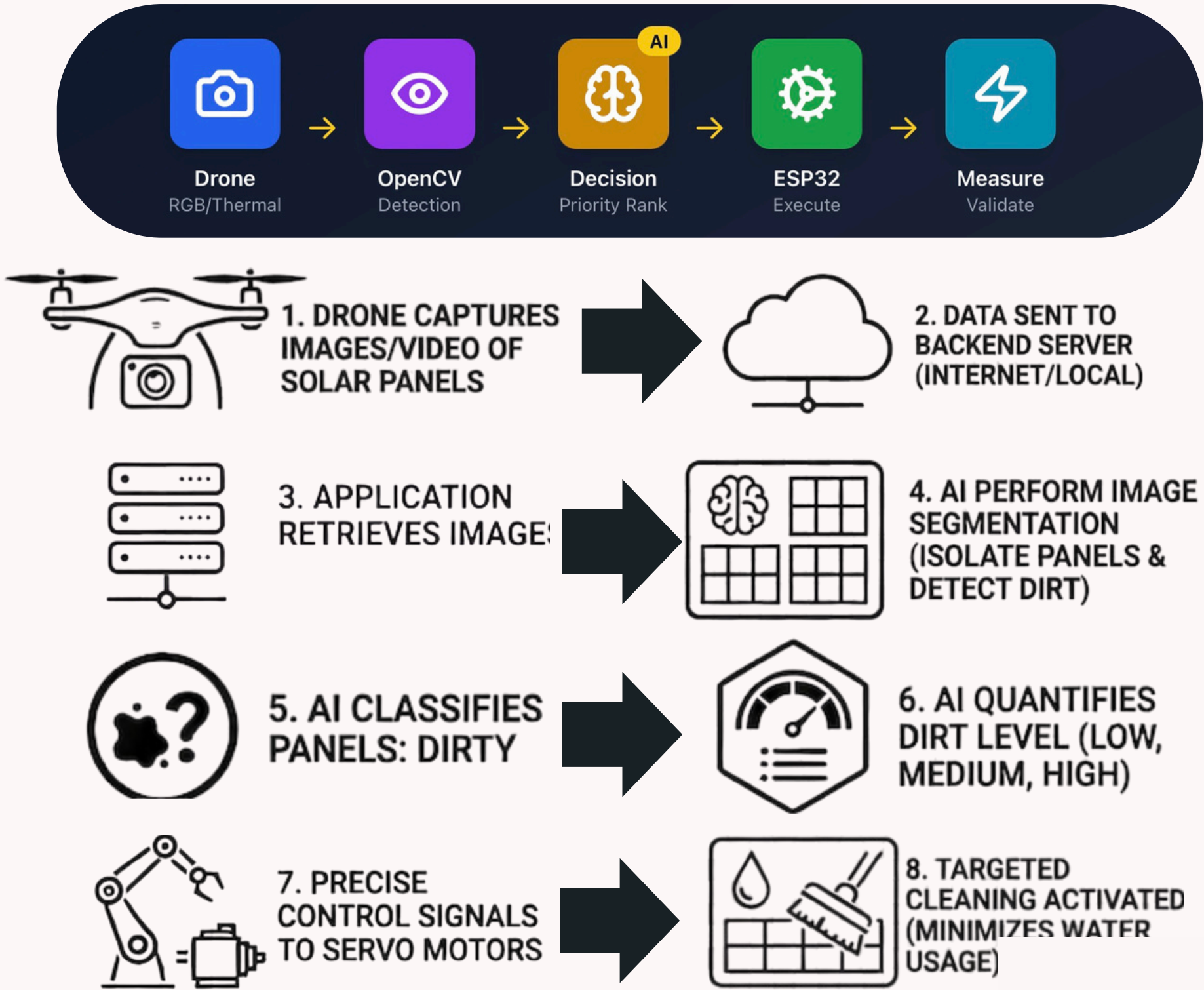


Servo operation time optimized at seconds per panel ($\approx 3-6$ s) instead of fixed-duration cleaning cycles, reducing mechanical wear.



Generates real-time ROI metrics (₹/day, ₹/month) by correlating energy recovered vs water and operational cost.

System Architecture



Implementation



Water Cleaning test

Average water consumption is 2.8L per panel & after agents are applied it is reduced to 800-900ml per panel. Cleaning time is reduced to 80-85%.



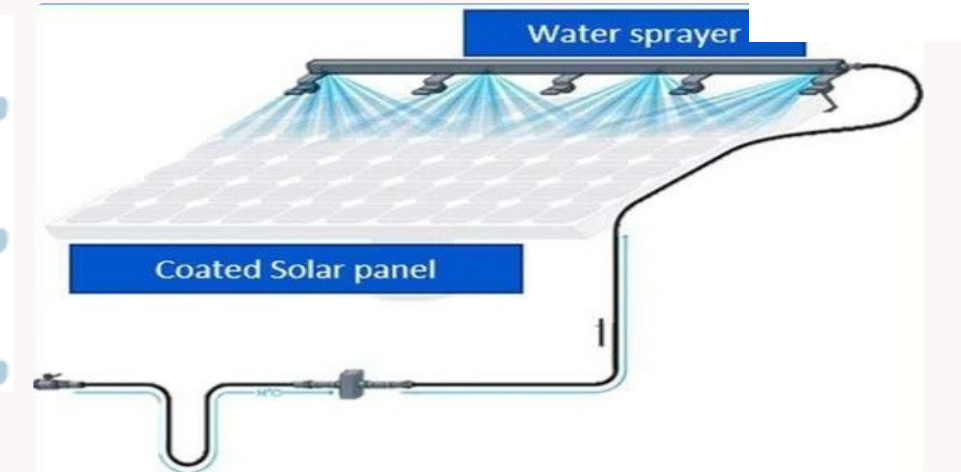
Electrical test

Electrical output is increased by 4.4% per panel after. Average cost saving 150rs per panel/month can be achieved.



Dust test

Due to dust accumulation power output is reduced up to 40%. Due to on the go cleaning automated panel settles 59.6% less dirt than traditional cleaning setups.



Agents

IoT Water Monitoring and Automation System



Confidence Agent:

Avoids wrong
cleaning



Priority Agent:

Ranks panels
by impact



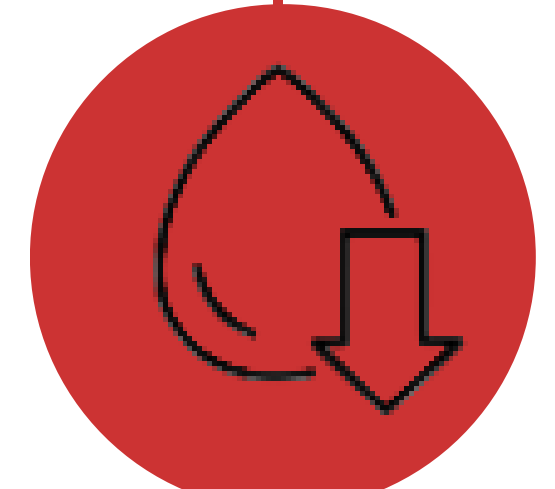
Benchmarking Agent:

AI vs traditional
cleaning



Impact & ROI Agent:

Calculates energy
& money gain



Water & Action Optimizer:

Decide servo and
pump

Dashboard

Overview

Live

Upload

Batch

Sessions

Analytics

Batch processing

Send multiple images at one run.

Drop images here or click to select

Process batch

Export CSV

Export PDF

6 file(s) selected

STATUS

Completed: 6/6 images

Preview

File

Class

Dir score

Time (ms)

imgclean_120_0.jpg

CLEAN

42.000

67

imgclean_926_0.jpg

CLEAN

35.000

58

imgclean_165_0.jpg

CLEAN

36.000

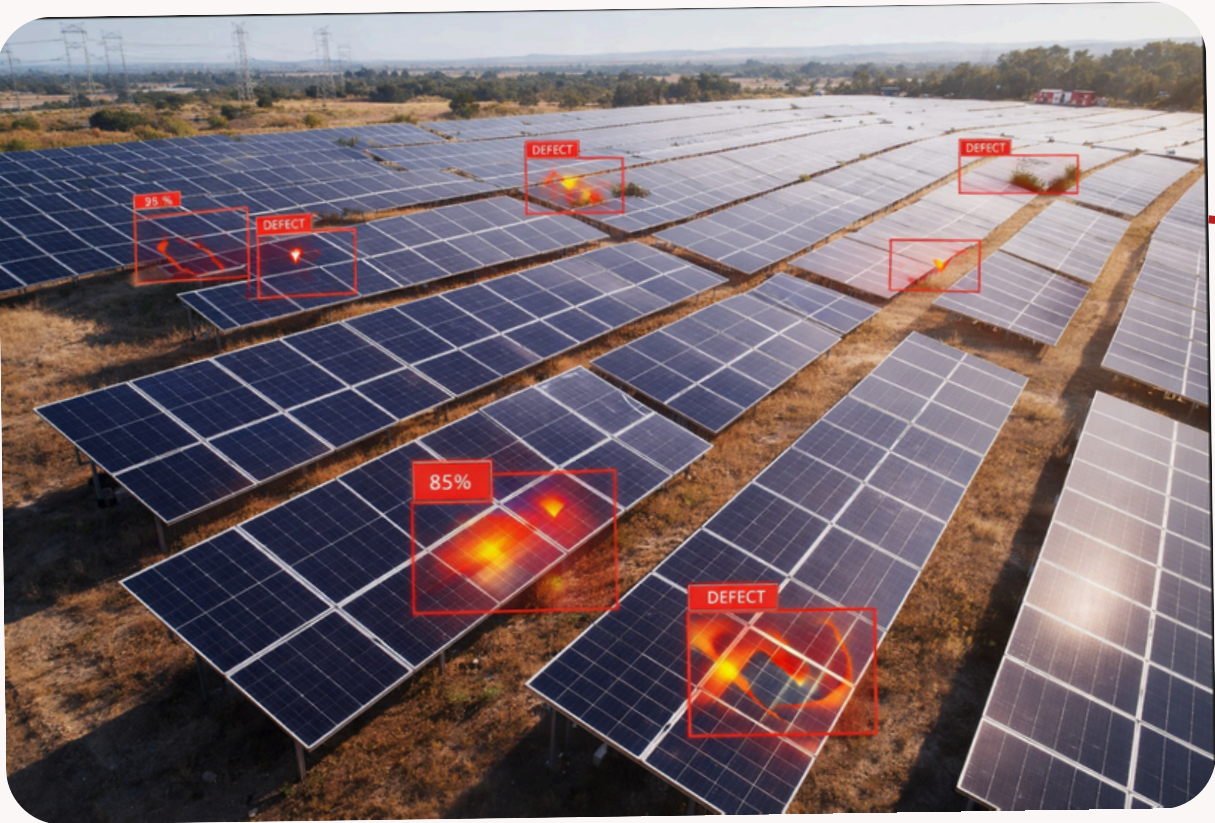
34

imgclean_1215_0.jpg

CLEAN

43.000

28



Impact Forecast

100

90

80

70

60

50

40

30

20

10

0

Current

After Cleaning

Traditional

AI Method

Water Efficiency

Energy Gain

Time Efficiency

Performance Comparison

Metric	Traditional	AI Method	Improvement
Water per Panel	2L	0.14L	+92.8%
Time per Panel	45s	5s	+87.3%
Efficiency Gain	35%	7.5%	+27.5%

Download Executive Report

Batch Analysis Results

4 panels analyzed

PANEL_20260118022140

19/100

Type: dust

Efficiency: +7.5%

Water: 0.14L

Savings: \$0.016/day

PANEL_20260118022141

67/100

Type: bird

Efficiency: +31.5%

Water: 1.02L

Savings: \$0.066/day

PANEL_20260118022141

33/100

Type: leaves

Efficiency: +14.5%

Water: 0.38L

Savings: \$0.03/day

PANEL_20260118022142

16/100

Type: leaves

Efficiency: +6%

Water: 0.18L

Savings: \$0.013/day

Download Batch Report

Clear Batch

19/100

DIRT SCORE

Light contamination

+7.5%

EFFICIENCY GAIN

0.131 kWh/day

\$0.016

DAILY SAVINGS

\$0.47/month

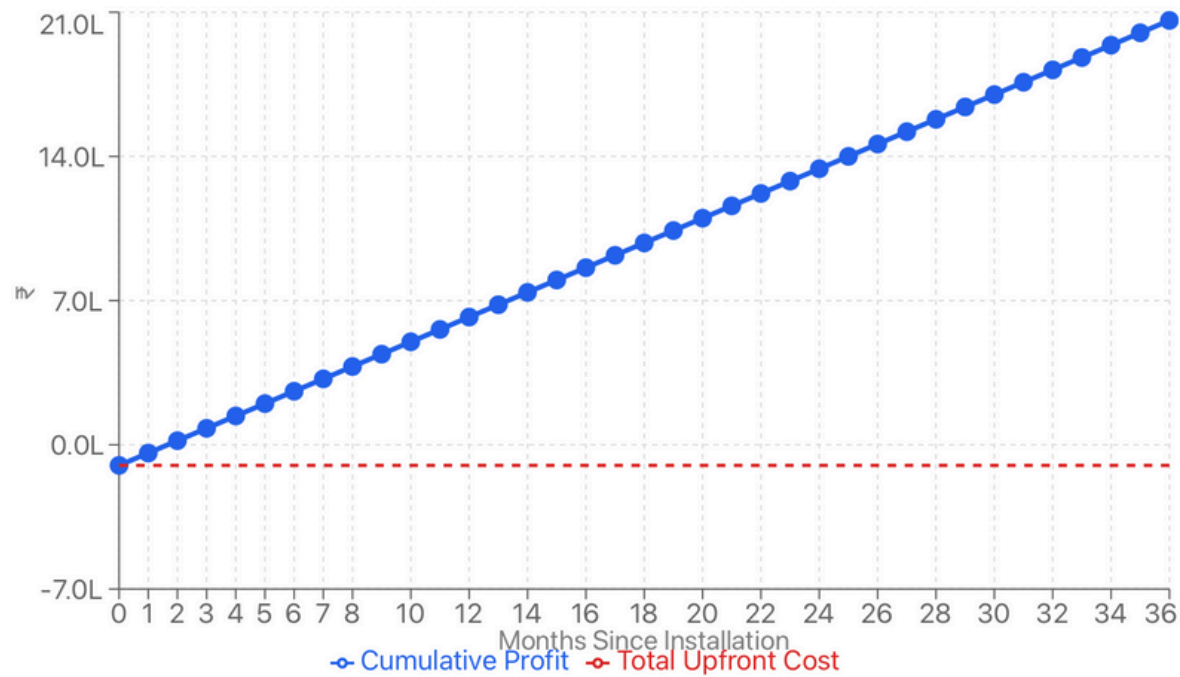
Computer Vision Analysis

AI Agent Decision

MONITORING ONLY

Unit Economics

Payback Timeline - 1 MW Solar Plant (₹1.0 Lakh Investment)



Monthly Savings

₹60,000

Break-even Point

2 months

ROI (3 years)

2160%

3 year Saving

₹20,60,000

1 MW Solar Plant Analysis

Traditional Manual Cleaning



Cost per kW per Month

₹80/kW

Total Monthly Cost (1 MW)

₹80,000

Annual Cost (1 MW)

₹9,60,000

Challenges: Multiple workers needed, weather dependent, inconsistent quality

Our Drone Technology



Cost per kW per Month

₹20/kW

75% cheaper than manual!

Initial Investment (1 Drone for 1 MW)

₹1,00,000

Only ₹100/kW one-time cost

Monthly Operating Cost (1 MW)

₹20,000

Benefits: Automated, consistent, scalable, minimal supervision

Cost Comparison Summary

Traditional Manual Cleaning

- Monthly Cost: ₹8,000
- Annual Cost: ₹96,000
- 3-Year Cost: ₹2,88,000
- Labor dependent, scheduling issues

Drone-Based Cleaning

- Initial Investment: ₹1,00,000
- Monthly Operating Cost: ₹2,000
- 3-Year Total Cost: ₹1,72,000
- Automated, consistent, scalable

Market Size

Borcelle, Co

\$1.41 billion

\$629 million

\$1.55M

TAM ● SAM ● SOM ●